months later Jenner inoculated the same boy with material taken from small pox patients. This was a dangerous but accepted procedure at that time. This procedure was called variolation. The boy was protected against small pox. His exposure to the mild cow pox disease had made him immune to the small pox disease. In this manner Jenner began the Science of Immunology, the study of the body's response to foreign substances. Edward Jenner was regarded as the **'Father of Immunology'**.



Figure 1.5: Dr. Edward Jenner performing his first vaccination (1796)

1.2.4 Robert Koch (1843-1910)

Robert Koch was a German physician and microbiologist (Figure 1.6). He was the founder of Modern Bacteriology. Robert Koch discovered *Bacillus anthracis* (Anthrax bacillus), *Mycobacterium tuberculosis*, and *Vibrio cholerae*. For the first time he showed the evidence that a specific germ (Anthrax bacillus) was the cause of a specific disease (splenic fever in sheep) and introduced scientific approach in Microbiology.

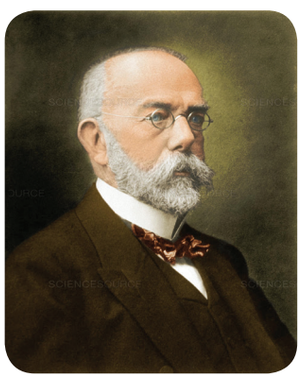


Figure 1.6: Robert Koch (1843-1910)

He modified Ziehl-Neelsen Acid Fast staining procedure which was introduced by Ehrlich. He devised solid medium to grow microorganism. He developed powerful method to isolate the microorganisms in pure culture from diseased tissue. He also perfected the techniques of identification of the isolated bacteria.

He introduced Koch's thread method to find out the efficacy of disinfectants. He established certain rules that must be followed to establish a cause and effect relationship between a microorganism and a disease. They are known as Koch's Postulates. He also described the Koch's Phenomenon. He was regarded as the **'Father of Medical Microbiology'**.

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Koch's Thread Method

Robert Koch carried out systematic experiments on disinfection, using pure cultures of bacteria. By means of his Thread Method, he investigated the effect on anthrax spores of the popular disinfectants at that time. Koch's Thread Method also called as carrier test. A carrier such as silk is contaminated by submerging in a liquid culture of the *Bacillus anthracis,* a test organism. The carrier is further dried and immersed in the disinfectant solution for a given exposure time. Thereafter the thread is cultured in a nutrient broth. No growth after incubation indicated that the product (disinfectant) is active.

Koch's Postulates

Four criteria were established by Robert Koch to identify the causative agents of an infectious disease. These include

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